Agent-based modeling as generative explanation

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Growing macro-patterns from micro-level cognitive rules

From controlled delegation to scientific modeling

In the previous lecture, the core problem was workflow control:

From controlled delegation to scientific modeling

In the previous lecture, the core problem was workflow control:

- define the goal

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In the previous lecture, the core problem was workflow control:

- define the goal
- constrain the implementation

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- verify the result

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ABM gives us a modeling problem where that discipline matters immediately.

Classical cognitive modeling

Cognitive models describe how one person:

Classical cognitive modeling

Cognitive models describe how one person:

- forms beliefs

Classical cognitive modeling

Cognitive models describe how one person:

- forms beliefs
- updates on evidence

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Cognitive models describe how one person:

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This is one standard level of analysis in cognitive science.

The gap

Many phenomena of interest are not properties of a single mind.

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- Why do populations polarize even when individuals reason well?

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- Why do norms emerge and persist without central enforcement?

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- Why do norms emerge and persist without central enforcement?
- Why do crowds sometimes find good solutions and sometimes fail?

A single-agent model can specify the cognitive mechanism, but it cannot by itself explain the population-level pattern.

An illustrative question

How can population-level polarization emerge even when individuals only make small, locally reasonable belief updates?

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We will treat this as a generative question, not a correlational one.

Why ABM?

- A standard cognitive model asks how one person updates.

Why ABM?

- A standard cognitive model asks how one person updates.
- An ABM asks what many interacting updaters produce together.

Why ABM?

- A standard cognitive model asks how one person updates.
- An ABM asks what many interacting updaters produce together.
- ABM embeds cognitive mechanisms inside an interaction structure.

What is an agent?

An agent is a simplified decision-maker with internal state, local inputs, and update rules.

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- State changes over time.

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An agent is a simplified decision-maker with internal state, local inputs, and update rules.

- State changes over time.
- Rules map local inputs to local actions.

Agent state examples

- belief

Agent state examples

- belief
- memory

Agent state examples

- belief
- memory
- attention

Agent state examples

- belief
- memory
- attention
- confidence

Agent state examples

- belief
- memory
- attention
- confidence
- strategy

Agent state examples

- belief
- memory
- attention
- confidence
- strategy
- preference

Agent state examples

- belief
- memory
- attention
- confidence
- strategy
- preference
- location

Agent state examples

- belief
- memory
- attention
- confidence
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- social ties

What is an agent-based model?

An agent-based model is a computational model in which possibly heterogeneous agents update their states through local rules while interacting with each other or with an environment.

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Components:

- agents and their states
- update rules
- environment and interaction structure
- update schedule
- macro-level outcomes

Intellectual lineage

- Schelling (1971): residential segregation from mild local preference

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In cognitive science, ABM connects naturally to social learning, cultural evolution, and language emergence.

ABM as generative explanation

To explain a macro-pattern, grow it from micro-level rules.

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1. Specify agents and their states.

ABM as generative explanation

To explain a macro-pattern, grow it from micro-level rules.

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ABM as generative explanation

To explain a macro-pattern, grow it from micro-level rules.

1. Specify agents and their states.
2. Specify interaction and update rules.
3. Run the system.

ABM as generative explanation

To explain a macro-pattern, grow it from micro-level rules.

1. Specify agents and their states.
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3. Run the system.
4. Check whether the target pattern appears.

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The explanatory claim is mechanism-level sufficiency.

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The explanatory claim is mechanism-level sufficiency.

But ABM does not license simple inverse inference: the same macro-pattern can arise from different micro-mechanisms.

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Emergence is the central explanandum of ABM.

ABM and statistical models answer different questions

- Statistical model: which variables predict observed outcomes?

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Good practice triangulates across all three.

Why cognitive scientists should care

- cultural transmission

Why cognitive scientists should care

- cultural transmission
- social learning

Why cognitive scientists should care

- cultural transmission
- social learning
- collective intelligence

Why cognitive scientists should care

- cultural transmission
- social learning
- collective intelligence
- polarization

Why cognitive scientists should care

- cultural transmission
- social learning
- collective intelligence
- polarization
- language emergence

Why cognitive scientists should care

- cultural transmission
- social learning
- collective intelligence
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Why cognitive scientists should care

- cultural transmission
- social learning
- collective intelligence
- polarization
- language emergence
- developmental systems
- cognitive ecology

What makes an ABM cognitive?

An ABM becomes a cognitive-science model when the agent rule represents a theory of cognition.

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- belief updating

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- belief updating
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- belief updating
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- decision thresholds

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The population model is only as cognitive as the agent model.

Toy model or empirical model?

Toy ABM:

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Toy ABM:

- asks whether a mechanism could generate a pattern

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- asks whether a mechanism could generate a pattern
- emphasizes conceptual clarity

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Empirical ABM:

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Here we will build a toy model explicitly.

Design checklist for ABM studies

1. target phenomenon

Design checklist for ABM studies

1. target phenomenon
2. agent-level mechanism

Design checklist for ABM studies

1. target phenomenon
2. agent-level mechanism
3. interaction mechanism

Design checklist for ABM studies

1. target phenomenon
2. agent-level mechanism
3. interaction mechanism
4. update schedule

Design checklist for ABM studies

1. target phenomenon
2. agent-level mechanism
3. interaction mechanism
4. update schedule
5. outcome measure

Design checklist for ABM studies

1. target phenomenon
2. agent-level mechanism
3. interaction mechanism
4. update schedule
5. outcome measure
6. sensitivity analysis

Design checklist for ABM studies

1. target phenomenon
2. agent-level mechanism
3. interaction mechanism
4. update schedule
5. outcome measure
6. sensitivity analysis
7. comparison model

Design checklist for ABM studies

1. target phenomenon
2. agent-level mechanism
3. interaction mechanism
4. update schedule
5. outcome measure
6. sensitivity analysis
7. comparison model
8. validation target

Running example: belief polarization

Research question:

Running example: belief polarization

Research question:

Can population-level polarization emerge from simple local social influence?

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Belief state: `b_i in [-1, 1]`

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- `-1`: strong belief in position A

Running example: belief polarization

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- `0`: neutral

Running example: belief polarization

Research question:

Can population-level polarization emerge from simple local social influence?

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- `-1`: strong belief in position A
- `0`: neutral
- `+1`: strong belief in position B

Environment and interaction structure

We use a ring network.

Environment and interaction structure

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- each agent has two neighbors

Environment and interaction structure

We use a ring network.

- each agent has two neighbors
- interaction is local

Environment and interaction structure

We use a ring network.

- each agent has two neighbors
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Environment and interaction structure

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- each agent has two neighbors
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The ring is deliberately unrealistic; it makes locality transparent.

Ring schematic

```
0 -- 1 -- 2 -- 3 -- ... -- N-1
|                          |
+--------------------------+
```

Ring schematic

```
0 -- 1 -- 2 -- 3 -- ... -- N-1
|                          |
+--------------------------+
```

Local edges constrain who can influence whom.

Update schedule

Synchronous: all agents update simultaneously each step.

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Asynchronous: one agent is chosen per step (random or sequential).

Update schedule

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Asynchronous: one agent is chosen per step (random or sequential).

The choice changes dynamics and is often under-reported – it is part of the model specification.

Cognitive updating rule

At each step:

Cognitive updating rule

At each step:

1. choose one focal agent

Cognitive updating rule

At each step:

1. choose one focal agent
2. choose one neighbor

Cognitive updating rule

At each step:

1. choose one focal agent
2. choose one neighbor
3. compare beliefs

Cognitive updating rule

At each step:

1. choose one focal agent
2. choose one neighbor
3. compare beliefs
4. assimilate if similar

Cognitive updating rule

At each step:

1. choose one focal agent
2. choose one neighbor
3. compare beliefs
4. assimilate if similar
5. contrast if dissimilar

Mathematical rule

```
if |b_i - b_j| < tau:
    b_i <- b_i + alpha * (b_j - b_i)
else:
    b_i <- b_i - beta  * (b_j - b_i)
clip b_i to [-1, 1]
```

Parameter interpretation

- `alpha`: assimilation rate

Parameter interpretation

- `alpha`: assimilation rate
- `beta`: contrast rate

Parameter interpretation

- `alpha`: assimilation rate
- `beta`: contrast rate
- `tau`: tolerance threshold

Parameter interpretation

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- `tau`: tolerance threshold

Parameter interpretation

- `alpha`: assimilation rate
- `beta`: contrast rate
- `tau`: tolerance threshold

These parameters define the cognitive-social mechanism.

Before coding, write the scientific contract

For this model, we need to specify:

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- research question

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- agent state variables

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For this model, we need to specify:

- research question
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- outcome metrics

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For this model, we need to specify:

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- expected qualitative behavior

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This is the same planning discipline we use with Cursor.

- initialize agents and beliefs

Python architecture

- initialize agents and beliefs
- define ring neighbors

Python architecture

- initialize agents and beliefs
- define ring neighbors
- implement `step()` update

Python architecture

- initialize agents and beliefs
- define ring neighbors
- implement `step()` update
- run simulation and record history

Python architecture

- initialize agents and beliefs
- define ring neighbors
- implement `step()` update
- run simulation and record history
- compute outcome measures

Minimal class outline

```python
class BeliefPolarizationABM:
    def __init__(...): ...
    def neighbors(self, i): ...
    def step(self): ...
    def run(self, ...): ...
```

Minimal step() excerpt

```python
difference = self.beliefs[j] - self.beliefs[i]
if abs(difference) < self.tolerance:
    self.beliefs[i] += self.alpha * difference
else:
    self.beliefs[i] -= self.beta * difference
self.beliefs[i] = np.clip(self.beliefs[i], -1, 1)
```

Implementation choices

- NetLogo: visual and educational; widely used in ABM

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- Mesa: Python ABM framework; integrates with NumPy, pandas, and matplotlib
- AgentPy: lightweight Python option; useful for parameter sweeps
- Plain Python: best here because every line of the mechanism stays visible

Outcome measures

- mean belief

Outcome measures

- mean belief
- variance

Outcome measures

- mean belief
- variance
- polarization: mean `|b_i|`

Outcome measures

- mean belief
- variance
- polarization: mean `|b_i|`
- extreme share: proportion near `-1` or `+1`

Outcome measures

- mean belief
- variance
- polarization: mean `|b_i|`
- extreme share: proportion near `-1` or `+1`
- optional cluster count

Verification checks for this model

Useful checks include:

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Useful checks include:

- beliefs always remain in [-1, 1]

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- no-interaction or no-contrast variants behave as expected

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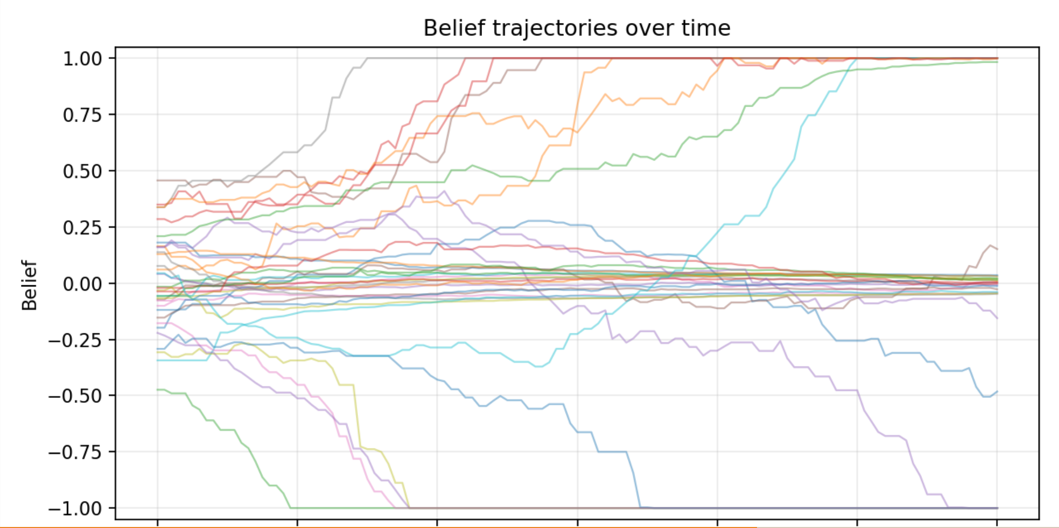
- output files are created in the expected folder

Verification checks for this model

Useful checks include:

- beliefs always remain in [-1, 1]
- the same seed reproduces the same trajectory
- no-interaction or no-contrast variants behave as expected
- output files are created in the expected folder
- parameter sweeps summarize multiple seeds, not one run

Individual trajectories



How to read trajectories

- agents start near the center

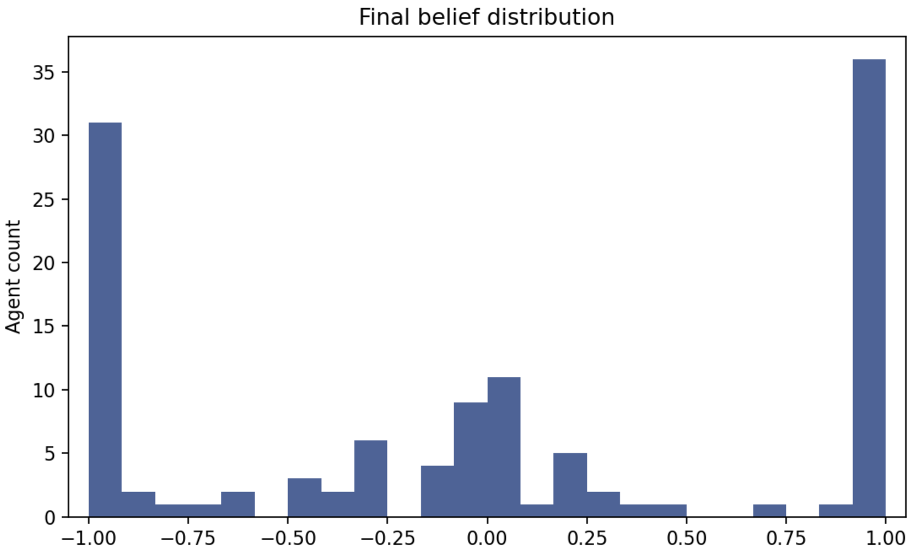
How to read trajectories

- agents start near the center
- local updates amplify differences

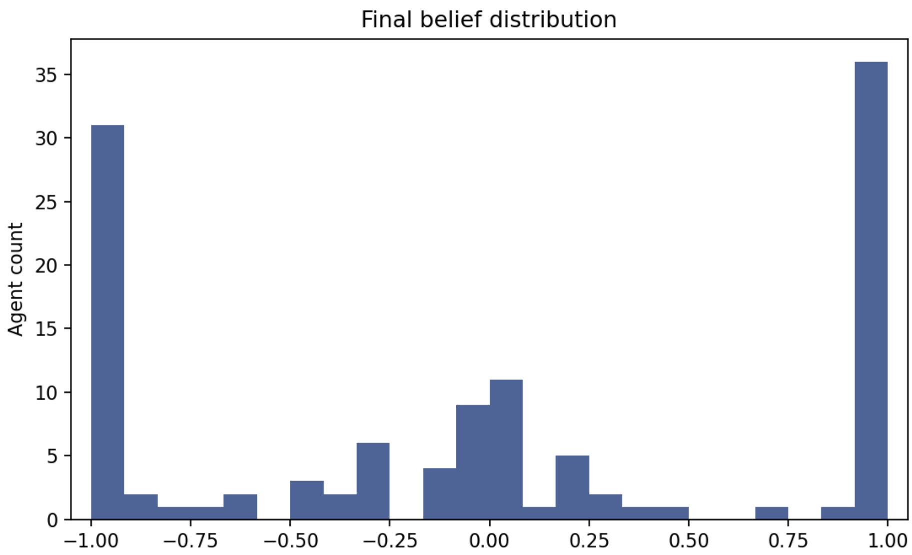
How to read trajectories

- agents start near the center
- local updates amplify differences
- repeated interaction can separate groups

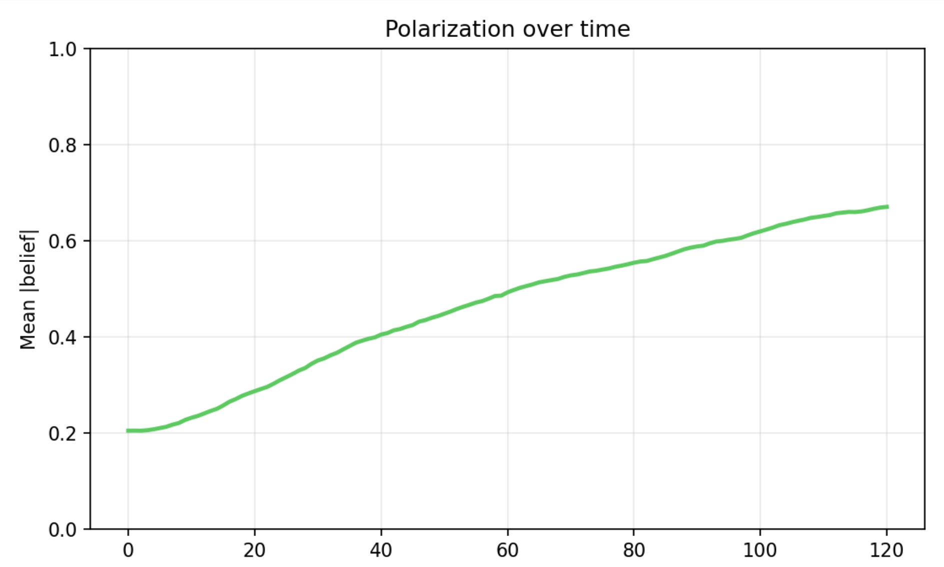
Final distribution



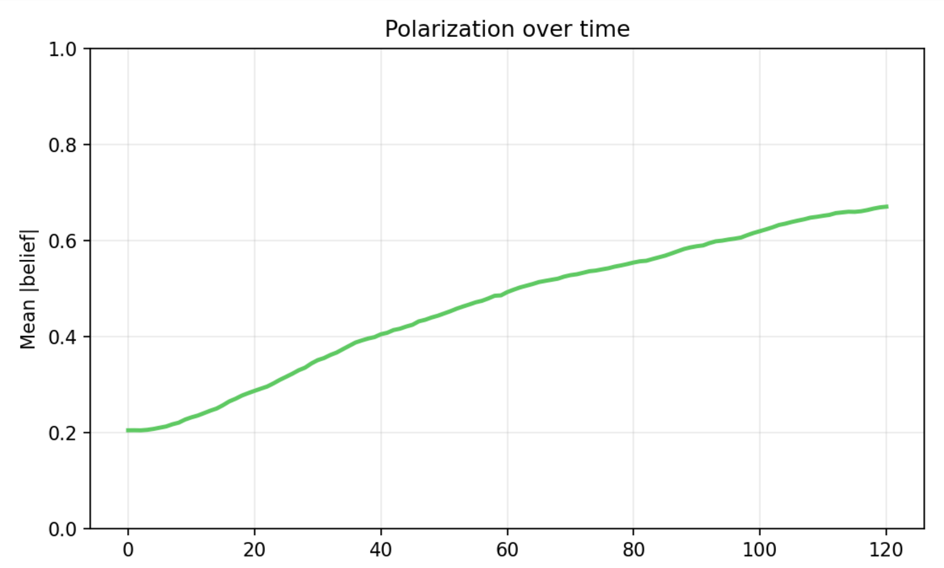
Final distribution



Polarization over time



Polarization over time



What emerged?

- No agent tries to polarize the population.

What emerged?

- No agent tries to polarize the population.
- Each agent applies a local update rule.

What emerged?

- No agent tries to polarize the population.
- Each agent applies a local update rule.
- The population distribution changes shape because local updates accumulate over time.

One run is not the result

ABMs often include stochastic elements:

One run is not the result

ABMs often include stochastic elements:

- initial beliefs

One run is not the result

ABMs often include stochastic elements:

- initial beliefs
- selected agents

One run is not the result

ABMs often include stochastic elements:

- initial beliefs
- selected agents
- selected neighbors

One run is not the result

ABMs often include stochastic elements:

- initial beliefs
- selected agents
- selected neighbors
- update order

One run is not the result

ABMs often include stochastic elements:

- initial beliefs
- selected agents
- selected neighbors
- update order
- random seeds

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ABMs often include stochastic elements:

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Analyze the distribution over runs, not one trajectory.

Sensitivity analysis is required

One run is a demonstration, not evidence of a robust mechanism.

Sensitivity analysis is required

One run is a demonstration, not evidence of a robust mechanism.

- vary tolerance

Sensitivity analysis is required

One run is a demonstration, not evidence of a robust mechanism.

- vary tolerance
- vary contrast strength

Sensitivity analysis is required

One run is a demonstration, not evidence of a robust mechanism.

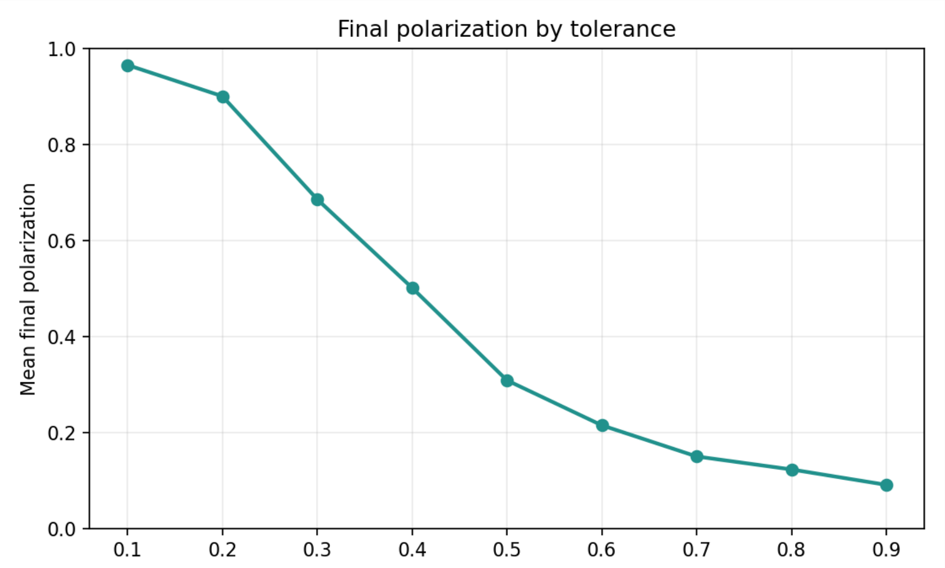
- vary tolerance
- vary contrast strength
- vary network structure

Sensitivity analysis is required

One run is a demonstration, not evidence of a robust mechanism.

- vary tolerance
- vary contrast strength
- vary network structure
- vary random seed

Parameter sweep over tolerance



Comparison models to run

- assimilation only (no contrast)

Comparison models to run

- assimilation only (no contrast)
- contrast only (no assimilation)

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What data constrain the model?

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- a simpler model produces the same pattern
- macro-patterns match but micro-dynamics are wrong
- the update rule contradicts empirical behavioral data

ABM in cognitive science

ABM does not replace cognitive modeling.

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ABM helps cognitive scientists ask whether a proposed cognitive mechanism, when embedded in an interaction structure, is sufficient to generate a population-level phenomenon.

Final assignment

You will build a small ABM that includes:

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ABM tests whether a proposed cognitive mechanism, embedded in an interaction structure, is sufficient to generate a target population-level phenomenon.

References

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